# Supplementary Information: Iterative Quantum Amplitude Estimation

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(Dated: February 1, 2021)

## 1. CANONICAL QAE WITH MLE

QAE can be enhanced by using a classical MLE post-processing. Given a parametrized probability distribution $f$ with unknown parameter $\theta$ and data $\{x_i\}_i$, $i = 1, \ldots, N$ sampled from it, MLE is a method to obtain an estimate $\hat{\theta}$ for $\theta$. This is done by maximizing the likelihood $L$,

$$\hat{\theta} = \arg\max_{\theta'} L(\theta') = \arg\max_{\theta'} \prod_{i=1}^{N} f(x_i|\theta'), \tag{1}$$

which is a measure for how likely it is to observe the data $\{x_i\}_i$, given that $\theta'$ is the true parameter. Numerically it is often favourable to maximize the log-likelihood $\log L$.

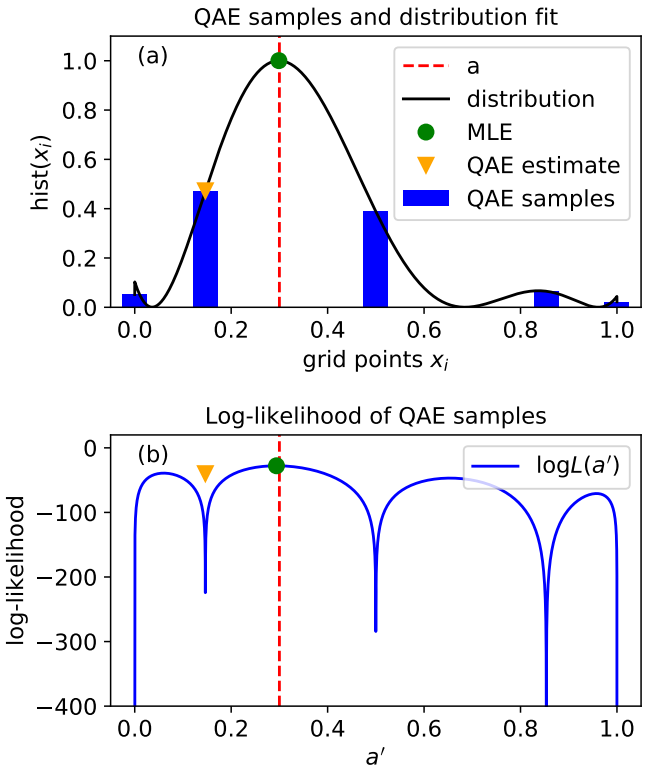
In QAE, we try to approximate $a$ with the MLE $\hat{a}$. The probability distribution $f$ is the probability to sample a certain grid point in one measurement. From [1], this distribution can be derived to be

$$f(x_i|a') = \begin{cases} \mathbb{P}[|y(a')\rangle] + \mathbb{P}[|M - y(a')\rangle], & \text{if } a' \notin \{0, 1\} \\ \mathbb{P}[|y(a')\rangle], & \text{otherwise} \end{cases}, \tag{2}$$

with $y(a') = M \arcsin(\sqrt{a'})/\pi$, $\mathbb{P}[|y\rangle]$ from Equation (3) from the main text, and grid points $x_i = \sin^2(i\pi/M), i \in \{0, \ldots, M/2\}$. See Supplementary Figure 1a for a visualization of the probability distribution fitted to the QAE samples and Supplementary Figure 1b for the respective log-likelihood function $\log L$.

To find the maximum of $\log L$ without much overhead, we exploit the information given by the QAE output: if QAE is successful its estimate is the closest grid point to $a$. Thus, we only have to search the intervals of the neighbouring grid points to find the exact $a$, which is done using a bisection search. Note that for $N \to \infty$, this search would return the exact amplitude, i.e. $\hat{a} = a$, independent of the number of qubits $m$.

Confidence intervals for the MLE can be derived using the Fisher information in combination with the central-limit theorem or with the likelihood ratio (LR) [2]. In our tests, the LR was more reliable than other approaches such as (observed) Fisher information. Due to the data-based definition of the LR confidence intervals, the better performance fits the expectations [3, 4]. The LR confidence interval uses the fact that for large sample numbers




SUPPLEMENTARY FIGURE 1: MLE postprocessing for QAE: In this problem, we set $a = 0.3$, $m = 3$ and used 25 shots for QAE. (a) The distribution (black line) is fitted to the the normalized histogram of the QAE samples (blue bars). The QAE output is the median of all samples (orange triangle). The MLE (green dot) in the QAE setting is also the peak of the distribution and very close to the real $a$. (b) The MLE is the global maximum of the log-likelihood function (blue line). The search for the MLE is conducted as a bisection search in the two neighbouring bubble-shaped intervals of the QAE estimate.

$N$ the LR statistic is approximately $\chi^2$-distributed, with one degree of freedom:

$$2 \log \frac{L(\hat{a})}{L(a)} \sim \chi^2_1. \tag{3}$$

The statistic can be used to conduct a two-sided hypothesis test with the null hypothesis $H_0 : \hat{a} = a$, which is rejected at the $\alpha$ level if the LR statistic

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exceeds the $(1-\alpha)$ quantile of the $\chi^2$ distribution, $q_{\chi^2_1}(1-\alpha)$. The corresponding confidence interval is $\left\{a' \in [0,1] : \log L(a') \geq \log L(\hat{a}) - q_{\chi^2_1}(1-\alpha)/2\right\}$. Using the likelihood function for the QAE samples we obtain confidence intervals for the MLE $\hat{a}$ of $a$. We apply the same approach to derive confidence intervals for MLAE.

## 2. PROOF OF THEOREM 1

Let us outline the strategy. We are going to use the union bound to combine the estimates, which are derived from the Chernoff-Hoeffding bound, for different rounds of the algorithm and an upper bound $T$ for the actual number of rounds $t$ required to derive an upper bound for the total query complexity in terms of the desired precision $\epsilon$ for the parameter $a$ and confidence level $1-\alpha$.

*Proof.* Suppose a given confidence interval $[a_l, a_u]$ for $a$, and recall that $a = \sin^2(\theta_a)$. This implies

$$\begin{aligned} \frac{|a_u - a_l|}{2} &= \frac{|\sin(\theta_u + \theta_l)||\sin(\theta_u - \theta_l)|}{2} \\ &\leq \frac{|\theta_u - \theta_l|}{2}. \end{aligned} \tag{4}$$

Thus, to achieve $|a_u - a_l|/2 \leq \epsilon$, it suffices to achieve $|\theta_u - \theta_l|/2 \leq \epsilon$ for our estimate of $\theta_a$.

Suppose that in round $i$ our knowledge of $\theta_a$ is the confidence interval $[\theta_l, \theta_u]$, and we just applied the Grover operator $k_i$ times. Recall that the application of $\mathcal{Q}^{k_i}$ to $\mathcal{A}|0\rangle$ effectively multiplies the angle $\theta_a$ by $K_i = 4k_i + 2$. Denote $\theta_i^{\min} := \{K_i\theta_l\}_{\mod 2\pi}$ and $\theta_i^{\max} := \{K_i\theta_u\}_{\mod 2\pi}$, where, unlike in Algorithm 2, we use fractional part modulo $2\pi$ of a given number. These are determined by measuring the variable $a_i$: their estimates $\tilde{a}_i$ for each round $i$ define the approximated probabilities of positive outcomes that we see from measurements and the corresponding confidence intervals $[a_i^{\min}, a_i^{\max}]$:

$$\begin{aligned} a_i &:= (1 - \cos(K_i\theta_a))/2, \\ a_i^{\min} &:= \max(0, \tilde{a}_i - \tilde{\epsilon}_{a_i}), \\ a_i^{\max} &:= \min(1, \tilde{a}_i + \tilde{\epsilon}_{a_i}), \\ \epsilon_{a_i} &:= \left|a_i^{\max} - a_i^{\min}\right|/2, \end{aligned} \tag{5}$$

where $\tilde{\epsilon}_{a_i}$ is a half-width of a confidence interval, calculated from Chernoff-Hoeffding bound for i.i.d. Bernoulli trials with $N_{\text{shots}}$ samples [5], which is given by

$$\mathbb{P}[a_i \notin [a_i^{\min}, a_i^{\max}]] \leq 2\exp\left(-2N_{\text{shots}}\tilde{\epsilon}_{a_i}^2\right). \tag{6}$$

Note, that actual error $\epsilon_{a_i}$ can be lower than $\tilde{\epsilon}_{a_i}$. If this scenario is realized, then $\theta_i^{\max} \in \{\pi, 2\pi\}$ or $\theta_i^{\min} \in \{0, \pi\}$, i.e. the confidence interval touches the boundary of the upper or lower half-circle.

Since we are looking for a sufficient number of measurement shots to ensure the algorithm runs correctly, we are going to refer to the number of shots as $N_{\max}$ from now on. If we require

$$2\exp\left(-2N_{\max}\tilde{\epsilon}_{a_i}^2\right) = \frac{\alpha}{T}, \quad \forall i \in \{1, \ldots, t\}, \tag{7}$$

then the union bound asserts that the main condition of the theorem – the guarantee for the total error probability being bounded by $\alpha$ – is satisfied:

$$\begin{aligned} &\mathbb{P}[a \notin [a_l, a_u]] \\ &\leq \mathbb{P}\left[\exists i \in \{1, \ldots, t\} : a_i \notin [a_i^{\min}, a_i^{\max}]\right] \\ &\leq \sum_{i=1}^{t} \mathbb{P}[a_i \notin [a_i^{\min}, a_i^{\max}]] \\ &\leq \sum_{i=1}^{t} 2\exp\left(-2N_{\max}\tilde{\epsilon}_{a_i}^2\right) = t\frac{\alpha}{T} \leq \alpha. \end{aligned} \tag{8}$$

From (7) and $\epsilon_{a_i} \leq \tilde{\epsilon}_{a_i}$ we deduce:

$$\epsilon_{a_i}^2 \leq \tilde{\epsilon}_{a_i}^2 = \frac{1}{2N_{\max}}\log\left(\frac{2T}{\alpha}\right), \quad \forall i \in \{1, \ldots, t\} \tag{9}$$

Now let us derive an upper bound $T$ for the actually required number of rounds $t$ to achieve the desired absolute error $\epsilon$ for $\theta_a$. First, given the actual schedule $\{q_i\}_{i=1}^{t-1}$, we have the following trivial relation for the last step of the algorithm:

$$\frac{L_{\min}}{K_t} \leq \epsilon < \frac{L_{\max}}{K_{t-1}}. \tag{10}$$

where $L_{\min}$ and $L_{\max}$ are lower and upper bounds for $\epsilon_{\theta_t}$ respectively (see Supplementary Equations (20)-(21)) and $K_t = K_1\prod_{i=1}^{t-1} q_i$. Moreover, the "no-overshooting" condition of Algorithm 1 gives us a stronger bound for $N_tK_t$:

$$N_tK_t \leq N_{\max}\frac{L_{\max}}{\epsilon}, \tag{11}$$

where $N_t$ stands for the number of shots at the last round of the algorithm. Secondly, we define an integer $T = T(r)$ for $r > 1$ as follows:

$$\frac{L_{\max}}{K_1 r^{T-1}} \leq \epsilon < \frac{L_{\max}}{K_1 r^{T-2}}. \tag{12}$$

i.e. $T := \lceil \log_r(rL_{\max}/2\epsilon)\rceil$ (we used $K_1 = 2$). Note, that this definition together with Supplementary Equations (10) and (17) leads to the following estimate:

$$r^{T-1-j} > \prod_{i=1}^{t-2-j} q_i, \quad \forall j \in \{0, \ldots, t-2\}, \tag{13}$$

which itself implies $r^{T-1} > r^{t-2}$, i.e $T \geq t$ and $T$ is an upper bound for $t$ as it should be.

If we define parameter $L \in [0, \pi/2]$ as

$$\sin^2(L) := 2\tilde{\epsilon}_{a_i}, \tag{14}$$

it follows from Supplementary Equation (9) that maximum number of shots required is given by

$$\begin{aligned} N_{\max}(\epsilon,\alpha) &= \frac{2}{\sin^4(L)}\log\left(\frac{2T(\epsilon)}{\alpha}\right) \\ &\leq \frac{2}{\sin^4(L)}\log\left(\frac{2}{\alpha}\log_r\left(\frac{r^2 L_{\max}}{2\epsilon}\right)\right). \end{aligned} \tag{15}$$

Lemma 1, which can be found below, finds the optimal value of $L$, denoted further in the proof by $L^*$, such that on every round $i$ we have the condition $\forall r \in (1,3]\ q_i \geq r$.

Finally, we derive a bound for the total number of $\mathcal{Q}$-oracle calls, denoted by $N_{\text{oracle}}$, using Supplementary Equations (11), (13) and (17):

$$\begin{aligned} N_{\text{oracle}} &= \sum_{i=1}^{t} N_i k_i = \sum_{i=1}^{t} N_i \frac{K_i - 2}{4} \\ &\leq \frac{N_{\max}}{2}\left(1 - t + \frac{K_1}{2}\sum_{i=1}^{t-2}\prod_{j=1}^{i} q_j\right) + \frac{N_t K_t}{2} \\ &< \frac{N_{\max}}{2}\left(1 - t + \sum_{i=1}^{t-2} r^{T-t+1+i} + \frac{L_{\max}}{2\epsilon}\right) \\ &= \frac{N_{\max}}{2}\left(1 - t + r^{T-t+1}\frac{r^{t-1}-r}{r-1} + \frac{L_{\max}}{2\epsilon}\right) \\ &= \frac{N_{\max}}{2}\left(r^{T-2}\frac{r^2}{r-1} + \frac{L_{\max}}{2\epsilon} + 1 - t - \frac{r^{T-t+2}}{r-1}\right) \\ &< \frac{N_{\max}}{2}\left(\frac{L_{\max}}{2\epsilon}\frac{r^2}{r-1} + \frac{L_{\max}}{2\epsilon} + 3 - t - \frac{r^{T-t+2}}{r-1}\right) \\ &< N_{\max}\frac{L_{\max}}{4\epsilon}\left(1 + \frac{r^2}{r-1}\right). \end{aligned} \tag{16}$$

Combining Supplementary Equations (15) and (16) with Lemma 1, we get

$$\begin{aligned} N_{\text{oracle}} &< \frac{1}{\epsilon} \times \frac{L_{\max}}{2\sin^4(L_{\max})}\left(1 + \frac{r^2}{r-1}\right) \\ &\quad \times \log\left(\frac{2}{\alpha}\log_r\left(\frac{r^2 L_{\max}}{2\epsilon}\right)\right), \end{aligned}$$

which, using $r = 2$, $L_{\max} = L^* < 11\pi/90 < \pi/8$ and $\sin^4(L_{\max}) = (1/4 - \sin(\pi/14)/2)^2$, gives us the bound

$$N_{\text{oracle}} < \frac{1}{\epsilon} \times \frac{44\pi}{9(1 - 2\sin(\pi/14))^2} \times \log\left(\frac{2}{\alpha}\log_2\left(\frac{\pi}{4\epsilon}\right)\right).$$

This inequality implies the following numerical form:

$$N_{\text{oracle}} < \frac{50}{\epsilon}\log\left(\frac{2}{\alpha}\log_2\left(\frac{\pi}{4\epsilon}\right)\right).$$

□

**Lemma 1** (Sufficient condition for $N_{\max}$)**.** For any $L \leq L^*$, where $L^* = \arcsin\left(\frac{1}{2}\sqrt{1 - 2\sin\left(\frac{\pi}{14}\right)}\right)$ the number of shots $N_{\max}(L)$, defined by Supplementary Equation (15), is sufficient to ensure

$$\forall r \in (1,3]\ q_i \geq r. \tag{17}$$

Moreover, $L_{\max} = L$ and $L_{\min} = \arcsin\big(\sin^2(L)\big)$.

*Proof.* Given a confidence interval $[\theta_i^{\min}, \theta_i^{\max}]$, it is equivalent to look at it as a pair of midpoint and error $\{\theta_i, \epsilon_{\theta_i}\}$, but we need to translate $\epsilon_{a_i}$ into $\epsilon_{\theta_i}$ first. Analogously to (4) we can write

$$\begin{aligned} \epsilon_{a_i} &= \frac{\big|\sin\big((\theta_i^{\max} + \theta_i^{\min})/2\big)\big|\big|\sin\big((\theta_i^{\max} - \theta_i^{\min})/2\big)\big|}{2} \\ &= \frac{|\sin(\theta_i)||\sin(\epsilon_{\theta_i})|}{2}, \end{aligned} \tag{18}$$

where $\theta_i := (\theta_i^{\max} + \theta_i^{\min})/2$. Now using Supplementary Equations (18) and (14) we can understand how to calculate the error $\epsilon_{\theta_i}$ from $\epsilon_{a_i}$ (or equivalently from $L$). For that purpose we introduce a family of functions $g_L(\theta_i)$

$$g_L(\theta_i) := \begin{cases} \min\left(\arcsin\left(\frac{\sin^2(L)}{\sin(\theta_i)}\right), \theta_i, \pi - \theta_i\right), & \theta_i \in \text{dom}^u(g_L) \\ \min\left(\arcsin\left(\frac{\sin^2(L)}{\sin(\theta_i - \pi)}\right), \theta_i - \pi, 2\pi - \theta_i\right), & \theta_i \in \text{dom}^d(g_L) \end{cases} \tag{19}$$

and we have $\epsilon_{\theta_i} = g_L(\theta_i)$, where the domain of $g_L$ depends on $L$ and equals $\text{dom}(g_L) = \text{dom}^u(g_L) \cup \text{dom}^d(g_L)$ with

$$\text{dom}^u(g_L) = \left[\arcsin\left(\frac{\sin(L)}{\sqrt{2}}\right), \pi - \arcsin\left(\frac{\sin(L)}{\sqrt{2}}\right)\right]$$

and

$$\text{dom}^d(g_L) = \left[\pi + \arcsin\left(\frac{\sin(L)}{\sqrt{2}}\right), 2\pi - \arcsin\left(\frac{\sin(L)}{\sqrt{2}}\right)\right]$$

Note, that the structure of domains reflects in itself the corner cases, when $\theta_i^{\max} \in \{\pi, 2\pi\}$ or $\theta_i^{\min} \in \{0, \pi\}$. Moreover, the function $g_L$ encodes nicely boundary cases when $\epsilon_{a_i} < \tilde{\epsilon}_{a_i}$ (i.e. when $2\epsilon_{a_i} < \sin^2(L)$). From the behavior of $g_L$ we also see that smallest possible error $\epsilon_{\theta_i}$ is attained at $\pi/2$ and equals

$$L_{\min} = \arcsin\big(\sin^2(L)\big). \tag{20}$$

Analogously,

$$L_{\max} = L \tag{21}$$

and $\epsilon_{\theta_i} = L_{\max}$ when $\theta_i = L$.

Observe now, that if there exists $j \in \{0, \ldots, 5\}$, such that

$$[\theta_i^{\min}, \theta_i^{\max}] \subseteq \left[\frac{j\pi}{3}, \frac{(j+1)\pi}{3}\right], \tag{22}$$

then we can always find a new $K_{i+1}$ with $q_i := K_{i+1}/K_i \geq 3$. Analogously, if there exists $j \in \{0, \ldots, 9\}$, such that

$$[\theta_i^{\min}, \theta_i^{\max}] \subseteq \left[\frac{j\pi}{5}, \frac{(j+1)\pi}{5}\right], \tag{23}$$

then we can always find a new $K_{i+1}$, such that $q_i \geq 5$. Finally, if there exists $j \in \{0, \ldots, 13\}$, such that

$$[\theta_i^{\min}, \theta_i^{\max}] \subseteq \left[\frac{j\pi}{7}, \frac{(j+1)\pi}{7}\right], \tag{24}$$

then we can always find a new $K_{i+1}$, such that $q_i \geq 7$. In all of the above cases, we have $q_i \geq 3$. In particular, the condition in Supplementary Equation (17) is satisfied.

In order to study when this happens we define piecewise linear functions $f_i$ on $[0, \pi]$ for every integer $i > 1$:

$$f_i(x) := \min\left(\tfrac{(j-1)\pi}{i} + x, \tfrac{j\pi}{i} - x\right)$$
$$\text{for every } j \in \{1, ..., i\} \text{ on every interval } \left[\tfrac{(j-1)\pi}{i}, \tfrac{j\pi}{i}\right] \tag{25}$$

On $[\pi, 2\pi]$ functions $f_i$ are defined analougosly by symmetry. These functions represent the distance from a given point on circle to the closest point from a set $\{\frac{j\pi}{i}\}_{j=0}^{2i}$. So we can consider functions $f_3, f_5, f_7$, which correspond to conditions in Supplementary Equations (22),(23) and (24) respectively. To ensure that at least one of this conditions is satisfied in a given round $i$, we can require that

$$g_L(\theta_i) \leq f_{\max}(\theta_i) \text{ for every } \theta_i \in \mathrm{dom}(g_L), \tag{26}$$

where $f_{\max}(\theta_i) := \max(f_3(\theta_i), f_5(\theta_i), f_7(\theta_i))$. The behavior of all functions is given in Supplementary Figure 2 for the top half of the circle - the bottom half is symmetric. Trivial analysis of these functions leads to an equivalent condition:

$$g_L\left(\frac{8\pi}{21}\right) \leq f_{\max}\left(\frac{8\pi}{21}\right) = \frac{\pi}{21} \tag{27}$$

and corresponding equation for $L^*$:

$$g_{L^*}\left(\frac{8\pi}{21}\right) = \frac{\pi}{21} \tag{28}$$

which gives us the formula

$$L^* = \arcsin\left(\frac{1}{2}\sqrt{1 - 2\sin\left(\frac{\pi}{14}\right)}\right). \tag{29}$$

□

**Remark 1.** We emphasize, that the proof provides an upper bound on the number of shots $N_{\max}$, which are sufficient to run the algorithm. This was essentially done to ensure that condition $q_i \geq r$ is satisfied, which itself allowed us to prove the upper bound. In practice, we can make less number of measurements $N_{\text{shots}} < N_{\max}$ in a given *iteration* until Algorithm 2 finds $q_i$ big enough to progress onto the next round. Because of that, a given *round* may consist of several iterations of the *while*-loop in Algorithm 1. Furthermore, we could get better constants in the proof if we use Clopper-Pearson interval bound instead of the Chernoff-Hoeffding bound. However, analytic treatment of this case is either impossible or highly involved, so we instead provide numerical evidence for the better performance of the Clopper-Pearson case in the main text, and a correctness Theorem 1 for a fixed range of precision parameter $\epsilon$ and level $\alpha$ via numerical method in Supplementary Section C.

## 3. A VARIANT OF THE THEOREM 1 FOR CLOPPER-PEARSON INTERVAL METHOD

Here we want to use Clopper-Pearson confidence interval [6, 7] instead of Chernoff-Hoeffding bound. Since it is impossible to obtain an analytic proof for this case, we will use a numerical calculations in the proof and fixed confidence level of 0.95, which is the most common one in practice.

**Theorem 1** (Correctness of IQAE with usage of Clopper-Pearson confidence interval)**.** Suppose a confidence level $1 - \alpha = 0.95$, a target accuracy $\epsilon \geq 2^{-200}$, and a number of shots $N_{\text{shots}} \in \{1, ..., N_{\max}(\epsilon, \alpha)\}$, where

$$N_{\max}(\epsilon, \alpha) = 69 \log\left(\frac{2}{\alpha}\log_2\left(\frac{\pi}{4\epsilon}\right)\right).$$

In this case, IQAE (Algorithm 1) terminates after a maximum number of $\lceil \log_2(\pi/8\epsilon) \rceil$ rounds, where we define one round as a set of iterations with the same $k_i$, and each round consists of at most $N_{\max}(\epsilon, \alpha)/N_{\text{shots}}$ iterations. IQAE computes $[\theta_l, \theta_u]$ with $\theta_u - \theta_l \leq 2\epsilon$ and

$$\mathbb{P}[\theta_a \notin [\theta_l, \theta_u]] \leq \alpha, \tag{30}$$

and returns $[a_l, a_u]$ with $a_u - a_l \leq 2\epsilon$ and

$$\mathbb{P}[a \notin [a_l, a_u]] \leq \alpha. \tag{31}$$

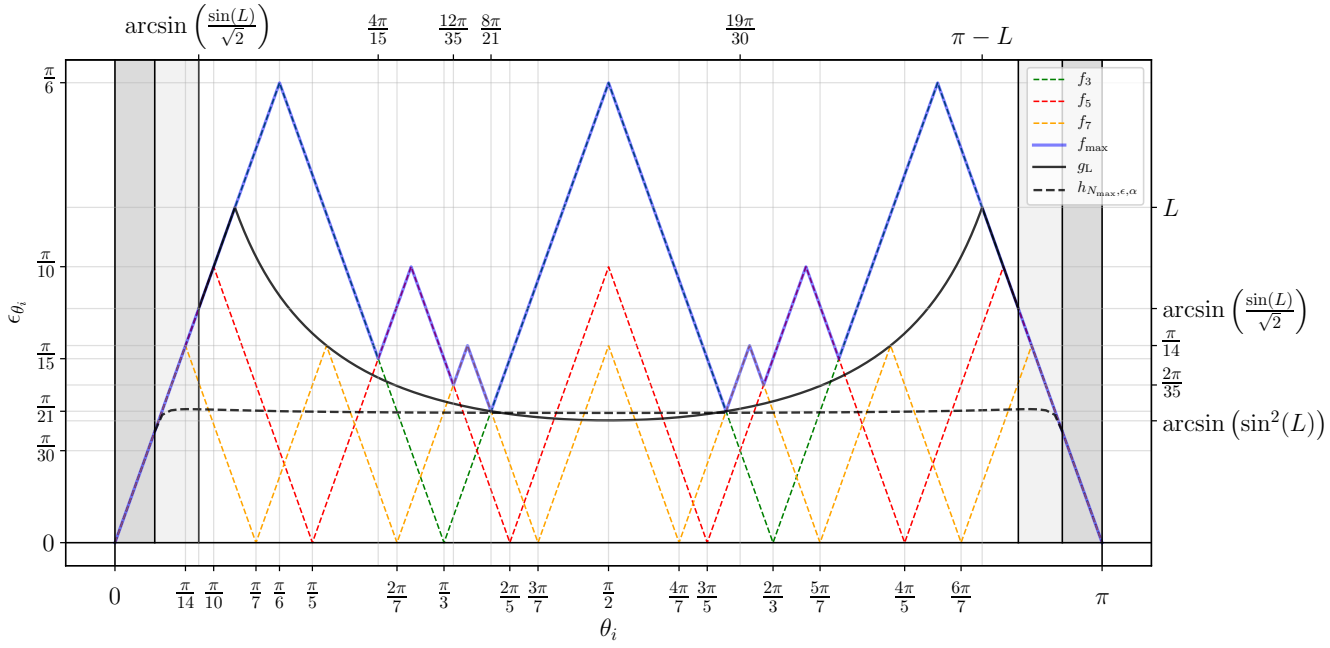
Thus, $\tilde{a} = (a_l + a_u)/2$ leads to an estimate for $a$ with $|a - \tilde{a}| \leq \epsilon$ with a confidence of $1 - \alpha$.

Furthermore, for the total number of $\mathcal{Q}$-applications, $N_{\text{oracle}}$, it holds that

$$N_{\text{oracle}} \;<\; \frac{14}{\epsilon}\log\left(\frac{2}{\alpha}\log_2\left(\frac{\pi}{4\epsilon}\right)\right) \tag{32}$$

*Proof.* The proof is completely analogous to Theorem 1 from the main text, except that instead of Chernoff-Hoeffding $1 - \alpha$ confidence interval we have different Clopper-Pearson condition for $[a^{\min}, a^{\max}]$:

$$\sup_a \mathbb{P}[a \notin [a^{\min}(X, n, \alpha), a^{\max}(X, n, \alpha)]] = \alpha \tag{33}$$




SUPPLEMENTARY FIGURE 2: Functions $f_3, f_5, f_7$ define the regions of validity for corresponding conditions (22),(23) and (24) respectively. The area under function $f_{\max}$ represents the logical union of these conditions. Finally, if we choose to use $N_{\max}(L)$ shots for the measurement, the function $g_L(\theta_i)$ represents the behavior of confidence interval width, which apparently depends on the estimated value $\theta_i$. Shaded regions are not included in the domain of $g_L(\theta_i)$, because the estimates $\theta_i$ cannot lie inside them for a chosen number of shots. The same is true for the case of Clopper-Pearson bound (see Supplementary Section C), which corresponds to the function $h_{N_{\max},\epsilon,\alpha}$.

where $X$ is Binomial random variable, constructed from $n$ Bernoulli trials with probability parameter $a$. This condition is satisfied for the following functions [7]:

$$a^{\min}(x,n,\alpha) := I^{-1}(1-\alpha/2; x+1, n-x) \quad (34)$$
$$a^{\max}(x,n,\alpha) := I^{-1}(\alpha/2; x, n-x+1) \quad (35)$$

where $I^{-1}(\alpha; x, y)$ is the inverse of regularized incomplete beta function, i.e. it is a solution $p$ of the equation $I_p(x,y) = \alpha$, where $I_p(x,y)$ is regularized incomplete beta function.

For each round Clopper-Pearson condition reads as

$$\sup_{a_i} \mathbb{P}\left[a_i \notin \left[a_i^{\min}, a_i^{\max}\right]\right] = \frac{\alpha}{T(\epsilon)} \quad (36)$$

where

$$a_i^{\min} := a^{\min}\left(X_i, N_{\max}, \frac{\alpha}{T(\epsilon)}\right)$$
$$a_i^{\max} := a^{\max}\left(X_i, N_{\max}, \frac{\alpha}{T(\epsilon)}\right)$$

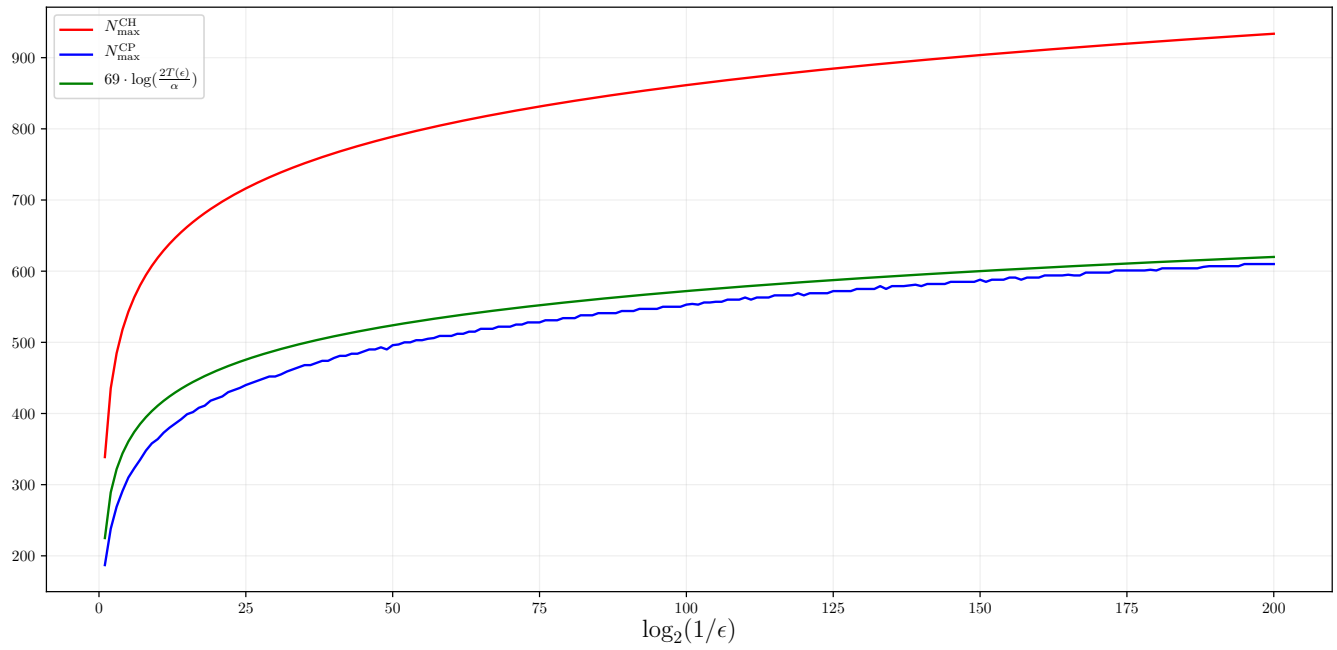
and $X_i$ are Binomial random variables for each round, representing the measured outcomes and $T(\epsilon)$ is given by Supplementary Equation (12).

In contrast to Chernoff-Hoeffding bound, this confidence interval depends on sampled value $x$ of Binomial random variable $X$, and it is no longer possible to introduce the parameter $L$ independent of $X$ to parametrize $N_{\max}$, as function of $\epsilon$ and $\alpha$, and find optimal value $L^*$ as was done in Lemma 1. Therefore we will find numerically the smallest sufficient number of shots $N_{\max}$, such that the conditions in Supplementary Equations (36) and (17) are satisfied for each round with a given confidence level $\alpha$ and precision $\epsilon$. In order to study that, we introduce in analogy to $g_L(\theta_i)$ the function $h_{N,\epsilon,\alpha}(\theta_i)$ defined in parametric way:

$$h_{N,\epsilon,\alpha}(\theta_i(t)) := \frac{1}{2}\arccos\left(1-2a^{\min}\left(t, N, \frac{\alpha}{T(\epsilon)}\right)\right) - \frac{1}{2}\arccos\left(1-2a^{\max}\left(t, N, \frac{\alpha}{T(\epsilon)}\right)\right)$$
$$\theta_i(t) := \frac{1}{2}\arccos\left(1-2a^{\min}\left(t, N, \frac{\alpha}{T(\epsilon)}\right)\right) + \frac{1}{2}\arccos\left(1-2a^{\max}\left(t, N, \frac{\alpha}{T(\epsilon)}\right)\right)$$

for $t \in \{0, 1, \cdots, N\}$. In order to find optimal number of shots $N_{\max}(\epsilon, \alpha)$, we need to solve the Supplementary Equation (28) for function $h_{N,\epsilon,\alpha}$:

$$h_{N_{\max}(\epsilon,\alpha),\epsilon,\alpha}\left(\frac{8\pi}{21}\right) = \frac{\pi}{21} \quad (37)$$




SUPPLEMENTARY FIGURE 3: Functions $N_{\max}^{\mathrm{CH}}$ and $N_{\max}^{\mathrm{CP}}$ are defined by Equation (6) from main text and Supplementary Equation (37) respectively for fixed $\alpha = 0.05$. They represent the smallest sufficient number of shots for a round, such that Supplementary Equation (17) is satisfied, and correspond to Chernoff-Hoeffding and Clopper-Pearson methods respectively. The green function Supplementary Equation (38) is a close approximation of the Clopper-Pearson exact blue result, which brings the bound to the same analytic form as in Chernoff-Hoeffding case.

We solve this numerically for fixed $\alpha = 0.05$ and $\epsilon \geq 2^{-200}$ and upper bound the result (see Supplementary Figure 3) with

$$N_{\max}(\epsilon, \alpha) = 69 \log\left(\frac{2}{\alpha} \log_2\left(\frac{\pi}{4\epsilon}\right)\right) \tag{38}$$

To get the corresponding bound for $N_{\text{oracle}}$ in accordance with Supplementary Equation (16) we need to calculate numerically $L_{\max}(\epsilon, \alpha)$, where

$$L_{\max}(\epsilon, \alpha) := \max_{\theta} h_{N_{\max}(\epsilon,\alpha),\epsilon,\alpha}(\theta). \tag{39}$$

We bound this function by a number $L_{\max}^* > L_{\max}(\epsilon, 0.05)$ for every $\epsilon$ in our range. Numerically we get $L_{\max}^* = 0.161$, which gives us the declared bound:

$$N_{\text{oracle}} < \frac{14}{\epsilon} \log\left(\frac{2}{\alpha} \log_2\left(\frac{\pi}{4\epsilon}\right)\right).$$

□


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